

UNITED STATES OF AMERICA
BEFORE THE FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the

Interstate Transmission System

Docket No. RM26-4-000

COMMENTS OF ANNESHA CHOWDHURY

I. Introduction and Statement of Interest

I respectfully submit these comments in Docket No. RM26-4-000 concerning the interconnection of large loads to the interstate transmission system. I am an artificial intelligence and data engineer specializing in AI systems for electric grid resilience and interconnection. I hold a Master of Science in Computer Science (Intelligent Systems) from the University of Texas at Dallas, am a Senior Member of the IEEE, and have approximately seven years of production engineering experience building large-scale data and AI systems at U.S. enterprises. I am the founder of Crisisense, an AI platform in development that applies physics-constrained machine learning to grid and critical-infrastructure decision support.

I file these comments in my individual capacity as a technologist working directly on the problem this proceeding addresses: how to integrate large new loads, including AI data centers, onto a constrained transmission system in a manner that is timely, orderly, reliable, and non-discriminatory. My comments focus on a specific, implementable opportunity that the Commission's record has not yet fully developed: the use of AI-assisted dynamic line rating and physics-constrained study automation to accelerate large-load interconnection without compromising reliability.

II. Summary of Recommendations

1. The Commission should recognize AI-assisted dynamic transmission line rating, built on the ambient-adjusted and dynamic rating foundation of Order No. 881, as a near-term tool for surfacing latent transmission headroom that can accommodate large loads without waiting for new line construction.
2. Any acceleration of interconnection or System Impact Study timelines toward the proposed 60-day target should be paired with a requirement that automated or AI-assisted study tools be validated against deterministic, physics-based engineering models, so that speed does not come at the cost of reliability.

3. The Commission should encourage standardized telemetry and data-exchange practices that make AI-assisted rating and study acceleration auditable and reproducible across transmission providers.

4. Flexible and curtailable large loads should be evaluated using real-time, consequence-aware analytics, which can quantify the reliability value of their flexibility and support the faster study track the ANOPR contemplates.

III. Discussion

A. AI-assisted dynamic line rating can surface existing headroom and reduce the need to defer large loads pending new construction.

A significant share of the interconnection delay facing large loads is not a physical absence of capacity but an inability to see and use capacity that already exists. Transmission lines are conventionally rated conservatively using static assumptions. Order No. 881's move toward ambient-adjusted ratings established that ratings can and should reflect actual conditions. Building on that foundation, AI methods that fuse real-time weather data with the IEEE 738 heat-balance model can identify dynamic headroom on existing lines, which can meaningfully exceed static ratings under favorable conditions. Where that latent headroom can be safely used, large loads can be integrated faster and at lower cost than approaches that assume new construction is always required. I recommend that the Commission expressly acknowledge AI-assisted dynamic line rating as an eligible tool in the large-load integration toolkit.

B. Faster study timelines must be coupled with physics-based validation to protect reliability.

The ANOPR asks whether interconnection studies could be completed within 60 days. Compressing study timelines is achievable with software and machine learning, but pure data-driven models can produce results that are statistically plausible yet physically invalid, which in grid operations can translate into reliability risk. The sound path is a hybrid, physics-constrained approach: machine-learning components accelerate data ingestion and scenario generation, while a deterministic engineering layer, such as power-flow and thermal models, validates every result against physical law before it is relied upon. I recommend that, to the extent the Commission encourages automated or AI-assisted study acceleration, it also direct that such tools incorporate a physics-based validation layer and remain auditable, so that faster studies remain trustworthy studies.

C. Flexibility and curtailment commitments should be evaluated with real-time, consequence-aware analytics.

The ANOPR contemplates a faster track for large loads that agree to be flexible and curtail usage. Determining how much reliability value a given flexibility commitment actually provides is itself an analytical problem well suited to real-time modeling. Consequence-aware analytics can quantify, for a specific location and system state, the headroom that a curtailable load frees up and the reliability margin its flexibility preserves. Embedding this kind of analysis in the faster study track would let the Commission and transmission providers extend expedited treatment on a principled, measurable basis rather than a categorical one.

D. Standardized, auditable data practices will make these tools usable across providers.

The benefits described above depend on consistent access to the underlying telemetry, including line conditions, weather inputs, and topology data. I recommend that the Commission encourage standardized data-exchange and documentation practices for any AI-assisted rating or study tools, so that their inputs and outputs are reproducible and auditable by transmission providers, regulators, and affected stakeholders. Standardization also lowers the barrier for smaller providers to adopt these methods.

IV. Conclusion

The need to integrate large loads, including AI data centers, in a timely and orderly manner is urgent, and tools to do so reliably are largely available today. AI-assisted dynamic line rating and physics-constrained study automation can help the Commission meet its timeliness goals while preserving the reliability and non-discrimination principles at the center of this proceeding. I appreciate the opportunity to comment and am available to provide further technical detail at the Commission's request.

Respectfully submitted,

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